

Sakib Mohammad, Ph.D.

Assistant Professor and Program Coordinator, Electronics Engineering Technology

Fairmont State University | U.S. Permanent Resident

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RESEARCH INTERESTS

Interpretable deep learning for microscopy and mechanobiology, computer vision with explainable AI, generative AI for biological data and materials science.

ACADEMIC APPOINTMENTS

Fairmont State University, Fairmont, WV

Assistant Professor, Electronics Engineering Technology

Aug 2025–Present

Program Coordinator:

Aug 2026–Present

- Teach undergraduate courses in circuits, electronics, digital systems, microcontrollers, embedded systems, robotics, and senior design.
- Lead program coordination, curriculum development, ABET accreditation activities, student mentoring for Electronics Engineering Technology.

Gopalganj Science and Technology University, Bangladesh

Lecturer, Department of Electrical and Electronic Engineering

Nov 2018–Jul 2019

- Delivered lectures and laboratory instruction, supervised undergraduate students, and contributed to departmental academic committees.

Green University of Bangladesh, Bangladesh

Adjunct Lecturer, Department of Electrical and Electronic Engineering

May 2018–Sep 2018

- Taught undergraduate courses and laboratories in circuits, signal processing, engineering drawing, and electrical service design.

GRADUATE RESEARCH AND TEACHING APPOINTMENTS

Southern Illinois University Carbondale, Carbondale, IL

Graduate Research Assistant, Mechanogenomics Laboratory

Aug 2021–Aug 2025

Southern Illinois University Carbondale, Carbondale, IL

Graduate Research Assistant, VLSI and RF Circuit Design Laboratory

Apr 2020–Jul 2021

Southern Illinois University Carbondale, Carbondale, IL

Graduate Teaching Assistant, Department of ECE and EET

Aug 2019–Jul 2024

EDUCATION

Southern Illinois University Carbondale, Carbondale, IL

Ph.D., Electrical and Computer Engineering; GPA: 4.00/4.00

Aug 2025

- Dissertation: *Deep Learning Powered Identification and Classification of Early Differentiated Germ Layers from Pluripotent Stem Cells with Interpretability*
- Advisor: Farhan H. Chowdhury

Manuscripts Under Review and Preprints

1. **Mohammad, S.**, & Hossain, M. S. (2026). Agentic Hydrogel: An agentic AI framework for hydrogel design, analysis and conditional generation. Manuscript under review at *Frontiers in Soft Matter – Gels*.
2. **Mohammad, S.**, Ritu, J., & Hossain, M. S. (2026). Single-channel tissue segmentation via cross-modal distillation from foundation models. *arXiv preprint arXiv:2606.00928*. Manuscript under review.
3. **Mohammad, S.**, Kausani, A. A., & Tousif, M. N. (2026). Evaluating image upsampling strategies for downstream microscopy image classification. *bioRxiv*. doi:10.1101/2026.02.13.705844. Manuscript under review.
4. Rathnayake, S., **Mohammad, S.**, Tragoudas, S., & Chowdhury, F. Deep learning-based evaluation framework for assessing the accuracy of traction force microscopy readouts. Manuscript under review at *IEEE Access*.

Peer-Reviewed Journal Articles

1. **Mohammad, S.**, Hossain, M. S., & Sarver, S. L. (2025). Integrating AI with cellular and mechanobiology: Trends and perspectives. *Biophysica*, 5, 62. doi:10.3390/biophysica5040062.
2. Roy, A., Sarver, S., Beights, J., Brennan, S., Rabi, S. N., **Mohammad, S.**, Han, K. Y., Nilufar, S., & Chowdhury, F. (2025). Cytoskeletal prestress regulates RIG-I-mediated innate immunity. *Biophysica*, 5(4), 51. doi:10.3390/biophysica5040051.
3. Hossain, M. S., Rabi, S. N., **Mohammad, S.**, Cook, K., Chowdhury, F., & Nilufar, S. (2025). Investigation of thermomechanical properties of hollow glass microballoon-filled composite materials developed by additive manufacturing with machine learning validation. *Polymers*, 17(11), 1495. doi:10.3390/polym17111495.
4. **Mohammad, S.**, Akand, R., Cook, K. M., Nilufar, S., & Chowdhury, F. (2024). Leveraging deep learning and generative AI for predicting rheological properties and material compositions of 3D printed polyacrylamide hydrogels. *Gels*, 10(10), 660. doi:10.3390/gels10100660.
5. **Mohammad, S.**, Roy, A., Karatzas, A., Sarver, S. L., Anagnostopoulos, I., & Chowdhury, F. (2024). Deep learning powered identification of differentiated early mesoderm cells from pluripotent stem cells. *Cells*, 13(6), 534. doi:10.3390/cells13060534.
6. **Mohammad, S.**, Amar, K., & Chowdhury, F. (2023). Hybrid AI models allow label-free identification and classification of pancreatic tumor repopulating cell population. *Biochemical and Biophysical Research Communications*, 677, 126–131. doi:10.1016/j.bbrc.2023.08.015.
7. Tousif, M. N., **Mohammad, S.**, Ferdous, A. A., & Hoque, M. A. (2018). Investigation of different materials as buffer layer in CZTS solar cells using SCAPS. *Journal of Clean Energy Technologies*, 6(4), 293–296. doi:10.18178/jocet.2018.6.4.477.

Peer-Reviewed Conference Papers

1. Tousif, M. N., Ushan, M. N. R., Akhter, R., Mahmud, S., Rashid, S., & **Mohammad, S.** (2024). Performance analysis of heart disease detection using different machine learning approaches. In *2024 6th International Conference on Electrical Engineering and Information & Communication Technology*. IEEE. doi:10.1109/ICEEICT62016.2024.10534366.
2. Tousif, M. N., Ushan, M. N. R., Joha, A. A., & **Mohammad, S.** (2017). A comprehensive study of CZTS solar cell simulation with ZnS buffer layer. In *2017 IEEE Region 10 Humanitarian Technology Conference* (pp. 193–197). IEEE. doi:10.1109/R10-HTC.2017.8288936.

3. **Mohammad, S.**, Tousif, M. N., Ferdous, A. A., Hoque, M. A., & Rahman, M. W. (2017). Numerical analysis of CZTS solar cell with In₂S₃ buffer layer. In *2017 IEEE International Conference on Smart Grid and Smart Cities* (pp. 55–60). IEEE. doi:10.1109/ICSGSC.2017.8038549.

Poster Presentations

1. **Mohammad, S.**, Rabi, S. N., & Chowdhury, F. (2025, April). Drug-induced morphological changes in HeLa cells: A deep learning and explainable AI perspective. Poster presented at CARP, Southern Illinois University Carbondale.
2. **Mohammad, S.**, Roy, A., Akand, R., & Chowdhury, F. (2024, April). Impact of various transfer learning methods on small biological dataset training. Poster presented at CARP, Southern Illinois University Carbondale.
3. **Mohammad, S.**, Amar, K., & Chowdhury, F. (2023, April). Tumorigenic pancreatic cancer cell classification with a combined deep and machine learning model. Poster presented at SRCAF, Southern Illinois University Carbondale.

GRANTS AND SPONSORED PROJECTS

Funded Projects

- **Edge-Based Object Detection for Search-and-Rescue Training Using Raspberry Pi.** University Faculty Fellowship Grant, Fairmont State University. **Role:** Principal Investigator. **Award:** \$5,000. 2026–2027
- **Growing Capacity Mini-Grant.** Appalachian Regional Commission / Fairmont State University. **Role:** Principal Investigator. **Award:** \$1,600. 2026

Selected Submitted Proposals

- **Interdisciplinary Discrete and Continuous Analysis in Appalachia.** National Science Foundation, Research Experiences for Undergraduates program. Niichel, R. J. (PI); Sedai, B. P. (Co-PI); Long, J. A.; Hossain, M.; MacLennan, B.; **Mohammad, S.** (Supporting Personnel); Sharma, G. **Amount requested:** \$372,000. *Not funded.* 2026
- **Generative Diffusion Modeling of Phase-Contrast Microscopy.** Falcon Mini-Grant, Fairmont State University. **Mohammad, S.** (Lead Applicant); Miller, J.; Kausani, A. A. **Amount requested:** \$8,000. *Not funded.* 2026

RESEARCH EXPERIENCE AND SELECTED CONTRIBUTIONS

- Developed interpretable deep-learning pipelines for phase-contrast and fluorescence microscopy, including classification, segmentation, cross-modal learning, and model interpretation with Grad-CAM, Grad-CAM++, LIME, and t-SNE.
- Trained classification and segmentation models for early mesoderm analysis, achieving 97% classification accuracy and 69% mIoU for phase-contrast images and 90% accuracy and 61% mIoU for nuclear images.
- Applied transfer learning to mechanobiology problems, including substrate-stiffness classification from phase-contrast microscopy (HeLa F1 = 0.92, B16F1 F1 = 0.80) and traction-force prediction using XGBoost and CNN-based models.
- Developed cross-modal distillation methods that transfer information from multimodal foundation models to single-channel tissue-segmentation models.
- Built predictive, generative, and agentic AI workflows for hydrogel data using MLPs, VAEs, conditional VAEs, normalizing flows, diffusion models, and multi-agent planning for material analysis and inverse design.
- Developed hybrid deep-learning and machine-learning pipelines for pancreatic tumor-repopulating cell classification and drug-induced morphological analysis in HeLa cells.
- Conducted microscopy acquisition using a Leica wide-field epifluorescence system, image processing in

MATLAB and ImageJ, traction-force microscopy workflows, and BSL-2 cell-culture work with murine embryonic stem cells and cancer cell lines.

- Lead an undergraduate edge-AI project using Raspberry Pi 5 and lightweight object-detection models for real-time search-and-rescue training.

TEACHING EXPERIENCE

Fairmont State University – Instructor of Record

Aug 2025–Present

- **Circuits and Electronics:** ELEC 1100 Circuit Analysis I, ELEC 1120 AC/DC Electronics Analysis, ELEC 2210 Circuit Analysis II, ELEC 3300 Advanced Linear Electronics.
- **Digital and Embedded Systems:** ELEC 2230 Digital Electronics, ELEC 2270 Introduction to Microcontrollers, ELEC 3310 Advanced Microcontrollers.
- **Capstone:** ELEC 4402 Senior Design II.
- Developed lectures, laboratories, assessments, and project-based activities involving circuits, and microcontrollers (Raspberry Pi, Arduino).

Southern Illinois University Carbondale – Graduate Teaching Assistant

Aug 2019–Jul 2024

- Supported laboratory instruction, circuit setup and debugging, grading, student consultation, and laboratory-manual preparation across introductory and advanced circuits, electronics, digital systems, and HDL courses.
- Courses included EET 238 Digital Systems Fundamentals, EET 245 Introductory Circuit Analysis and Applications, EET 245L Introductory Circuit Analysis and Applications Lab, EET 304A AC/DC Circuit Theory and Application, EET 304B Network Theory and Application; EET 403A Electronic Circuit Analysis, EET 403B Electronics Application and Design, ECE 235 Electrical Circuits I, ECE 235L Electrical Circuit I Lab, ECE 327 Digital Circuit Design with HDL, ECE 327L Digital Circuit Design with HDL Lab.

Gopalganj Science and Technology University – Lecturer

Nov 2018–Jul 2019

- Courses: EEE 321 Power Electronics, EEE 322 Power Electronics Laboratory, EEE 102 Electrical Circuits I Laboratory.

Green University of Bangladesh – Adjunct Lecturer

May 2018–Sep 2018

- Courses: EEE 101 Electrical Circuits I, EEE 406 Discrete Signal Processing Laboratory, EEE 340 Engineering Drawing and Electrical Service Design Laboratory.

STUDENT MENTORING

- Undergraduate Research Mentor, Edge AI and Raspberry Pi Object Detection Project, Fairmont State University 2026–Present
- Graduate and Undergraduate Student Mentor, Mechanogenomics Laboratory, Southern Illinois University Carbondale 2022–2025
- NSF REU Student Mentor, Mechanogenomics Laboratory, Southern Illinois University Carbondale 2022
- Volunteer Mentor, K-12 students in Python, data science, and machine-learning fundamentals 2023–2024

ACADEMIC LEADERSHIP AND INSTITUTIONAL SERVICE

- Program Coordinator, Electronics Engineering Technology, Fairmont State University Aug 2026–Present
- Member, Institutional Review Board, Fairmont State University Aug 2026–Present
- Contributor to ABET accreditation, program assessment, documentation, and continuous-improvement activities for Electronics Engineering Technology 2025–Present

- Lead and contribute to curriculum development and modernization, including development of new courses and revision of existing courses in electronics, circuits, and embedded systems 2025–Present
- Faculty Mentor, Facotronix Robotics Club (VEX U competitions), Fairmont State University 2025–Present
- Contributor, Industrial Advisory Committee meetings and regional industry engagement, Fairmont State University 2025–Present
- Co-organizer, Industrial Visit to Leer Mine, Fairmont State University 2025
- Member, Examination Moderation Committee, Gopalganj Science and Technology University 2018–2019

PROFESSIONAL SERVICE

Journal Peer Review

- *Cell Biochemistry and Biophysics* (Springer Nature) 2026–Present
- *Scientific Reports* (Nature Portfolio) 2025 – Present
- *Health Technology* (AME Publishing Company) 2025 – Present
- *Exploration of Medicine* (Open Exploration Publishing Inc.) 2025 – Present

Conference Service

- Technical Program Committee Member, International Conference on Machine Learning and Applications (ICMLA) 2025, 2026

STEM OUTREACH AND COMMUNITY ENGAGEMENT

- Instructor, STEMPact Summer Camp, Fairmont State University Jun 2026
- Microcontroller Demonstration Instructor, WV Space Grant Consortium and WV High Technology Foundation Apr 2026
- Electronics Engineering Technology Representative, Maroon and White Day, Fairmont State University Apr 2026
- Representative, West Virginia Marion County STEAM and CTE Festival Mar 2026
- Judge, VEX V5 Robotics Competition, WV Robotics Alliance Mar 2026
- Judge, 2026 Region VII Science and Engineering Fair, Fairmont State University Feb 2026

HONORS AND AWARDS

- Dissertation Research Assistantship 2024, 2025
- Doctoral Fellowship 2024
- Bangladesh Sweden Trust Fund Scholarship (Government Travel Grant) 2021
- OIC Scholarship (Undergraduate Entrance Merit Award) 2012

PROFESSIONAL DEVELOPMENT

- Faculty development workshops and instructional training, Center for Teaching Excellence, Fairmont State University 2025, 2026
- Blackboard Ultra and instructional-technology training 2025

TECHNICAL SKILLS

Machine Learning and Computer Vision: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, Hugging Face Transformers; CNNs, vision transformers, segmentation, knowledge distillation, generative models, explainable AI.

Scientific Computing and Data: Python, MATLAB, NumPy, Pandas, SciPy, SQL, Jupyter Notebook; Matplotlib, Seaborn, Plotly, TensorBoard, Tableau.

Embedded Systems and Prototyping: Raspberry Pi, Arduino, C/C++, Python, edge AI.

Cloud and Research Prototyping: Google Colab, Gradio, Streamlit, Hugging Face Spaces, Git.

Circuit and EDA Tools: Cadence Virtuoso, Verilog, PSpice, LTspice, Proteus, Synopsys Sentaurus.

Imaging and Laboratory Tools: ImageJ, Leica LAS X, epifluorescence microscopy, traction-force microscopy, BSL-2 cell culture.

REFERENCES

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